ORGNZ-04: Permissions in Grail Overview

> **Series:** ORGNZ — Organize Data: Buckets, Segments, Security | **Notebook:** 4 of 10 | **Created:** January 2026 | **Last Updated:** 04/25/2026

Overview

Dynatrace provides a comprehensive permission model for Grail that applies to all telemetry data including metrics, logs, spans, and events. Permissions can be assigned at the **bucket**, **table**, **record**, and **field** level. Without permissions, users cannot query data from Grail.

Prerequisites

Requirement	Details
Dynatrace Environment	SaaS environment with Grail enabled
Permissions	Account admin or IAM policy management access
Knowledge	Completed ORGNZ-03 (Bucket Strategy)
Data	None required (conceptual overview)

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Dynatrace Account	Account-level administrative access
Permissions	IAM policy management permissions
Knowledge	Completed ORGNZ-01 through ORGNZ-03

Learning Objectives

By the end of this notebook, you will:
- Understand the four permission levels in Grail
- Know how IAM policies control data access
- Recognize supported conditions for access control
- Navigate policy management in Dynatrace

Permission Levels

Grail supports permissions at multiple granularity levels:

Level	Granularity	Example Use Case
Bucket	All records in a bucket	Team owns entire bucket
Table	All records of a data type	Access to all logs across buckets
Record	Individual records by attribute	Filter by host group, namespace, security context
Field	Specific fields on records	Mask sensitive fields

IAM Policy Structure

Permissions are defined through IAM policies using statement queries:

```
ALLOW &lt;service&gt;:&lt;resource&gt;:&lt;action&gt; WHERE
&lt;conditions&gt;
```

Policy Components

Component	Description	Examples
Service	The Dynatrace service	`storage`
Resource	What is being accessed	`buckets`, `logs`, `metrics`, `spans`
Action	What operation is allowed	`read`, `write`, `delete`
Conditions	Contextual requirements	`bucket-name`, `security-context`

Supported Conditions for Storage Policies

Deployment-Level Attributes

Condition	Description	Example
`storage:bucket-name`	Filter by bucket name	`WHERE storage:bucket-name = 'team_logs'`
`storage:k8s.namespace.name`	Kubernetes namespace	`WHERE storage:k8s.namespace.name = 'production'`
`storage:k8s.cluster.name`	Kubernetes cluster	`WHERE storage:k8s.cluster.name = 'main-cluster'`
`storage:host.name`	Host name	`WHERE storage:host.name = 'web-server-01'`
`storage:dt.host_group.id`	Host group identifier	`WHERE storage:dt.host_group.id STARTSWITH 'prod-'`

Cloud Provider Attributes

Condition	Description
`storage:aws.account.id`	AWS account ID
`storage:gcp.project.id`	GCP project ID
`storage:azure.subscription`	Azure subscription
`storage:azure.resource.group`	Azure resource group

Custom Security Context

Condition	Description
`storage:dt.security_context`	Custom security context field

Policy Operators

IAM policies support several operators for conditions:

Operator	Usage	Example
`=`	Exact equality	`storage:bucket-name = 'team_logs'`
`STARTSWITH`	Prefix matching	`storage:bucket-name STARTSWITH 'prod_'`
`IN`	List equality	`storage:bucket-name IN ('bucket_a', 'bucket_b')`
`MATCH`	Pattern matching	`storage:bucket-name MATCH ('*-database-*')`

> **Important**: When the field holds an array (like `dt.security_context`), use `MATCH` instead of `=`, `STARTSWITH`, or `IN`. The latter operators will always return false for array fields.

Basic Policy Examples

Example 1: Default Bucket Access

Grant read access to all default buckets:

```
ALLOW storage:buckets:read WHERE storage:bucket-name STARTSWITH "default_";
ALLOW storage:events:read, storage:logs:read, storage:metrics:read,
      storage:entities:read, storage:bizevents:read, storage:spans:read;
```

Example 2: Team-Specific Bucket Access

Restrict access to team buckets:

```
ALLOW storage:buckets:read WHERE storage:bucket-name STARTSWITH
"team_platform_";
ALLOW storage:logs:read;
```

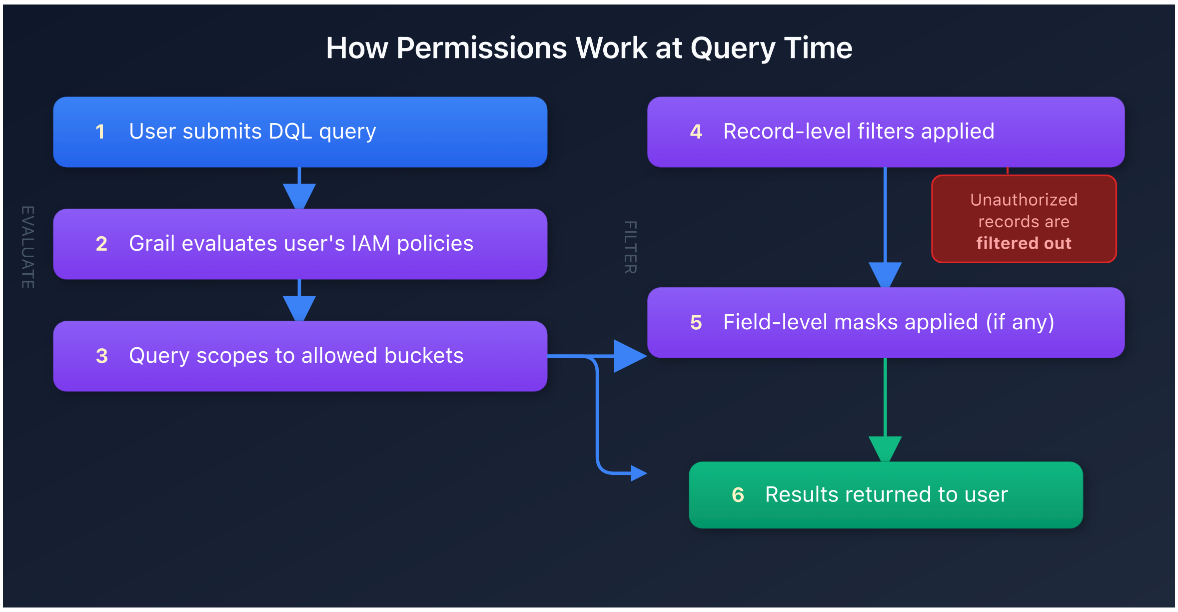
Example 3: Namespace-Based Access

Grant access based on Kubernetes namespace:

```
ALLOW storage:buckets:read WHERE storage:bucket-name STARTSWITH "default_";
ALLOW storage:logs:read WHERE storage:k8s.namespace.name = "production";
```

How Permissions Work at Query Time

When a user runs a DQL query, Grail evaluates permissions at each level:



Records without defined access are automatically filtered out.

Policy Management Location

To configure permissions:

1. Go to **Account Management**
2. If you have multiple accounts, select the appropriate account
3. Navigate to **Identity & access management** > **Policy management**

Policy Assignment

Target	Description
Groups	Assign policies to user groups (recommended)
Users	Assign policies to individual users
Service accounts	Assign policies for API access

Permission Scalability

Challenge

Creating access policies solely on the bucket and table level is not scalable in enterprise environments:

- One Dynatrace tenant has a limited number of custom buckets (80 default)
- Creating a bucket per team/application doesn't scale

Solution

Use record-level permissions based on:

Approach	Method
Deployment attributes	Host groups, namespaces, clusters
Cloud attributes	AWS account, GCP project, Azure subscription
Custom security context	`dt.security_context` field

This allows multiple teams to share buckets while seeing only authorized data.

Choosing Your Permission Strategy

Scenario	Recommended Approach
Small org, few teams	Bucket-level policies

Scenario	Recommended Approach
Large org, many teams	Security context + record-level
Kubernetes-centric	Namespace-based policies
Multi-cloud	Cloud account-based policies
Complex hierarchy	Hierarchical security context

Next Steps

Continue with the ORGNZ series:
- **ORGNZ-05**: Bucket-Level Access Control
- **ORGNZ-06**: Security Context
- **ORGNZ-07**: Advanced Permission Patterns

References

- Permissions in Grail
- IAM policy reference
- Working with policies

This notebook was AI-generated from Dynatrace documentation and enterprise best practices. It is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.